Machine instr. characteristics

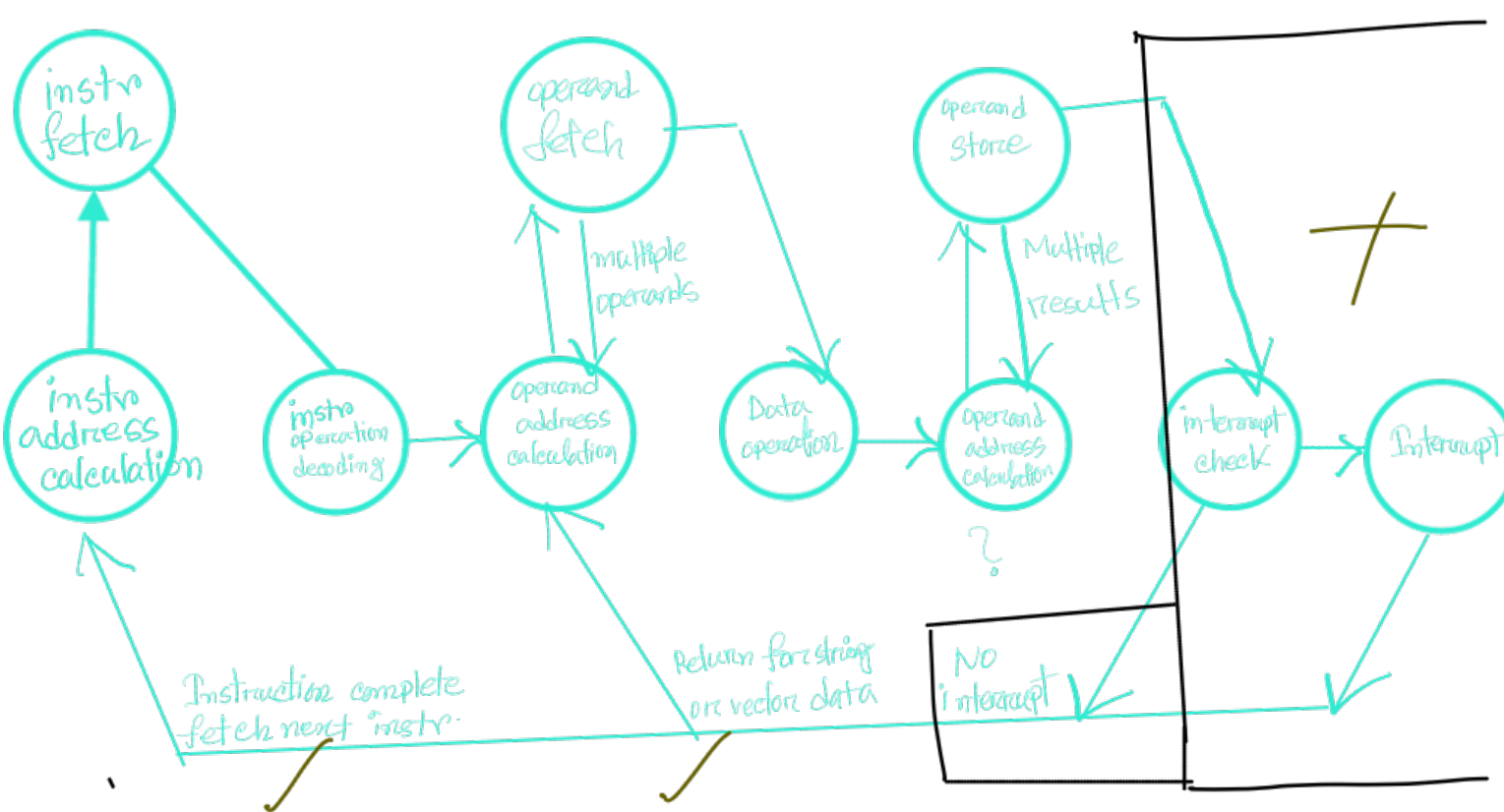
Machine instr set → Functional requirements for the processor.

program in ML → know abt register & Mem structure

→ Data type Machine directly supports

→ ALU's Function

processor's instr set: collection of different instructions the processor can execute.



Elements of MI:

1. Opcode: specifies action (ADD, LOAD) → a binary code.

2. Source Operand Reference: inputs needed for operation.

3. Result Operand Reference: Destination for output.

4. Next instr Reference: next instr. location.

↳ Real vs virtual:

(physical)

ISA distincts.

Implicit vs explicit:

↓

next immediately follows the current one.

→ instr. must include address from main/ virtual memory when needed.

Operand Sources

1. Main/virtual Mem: instr. must specify the mem address.
2. Processor Registers: → 1 reg: implicit
   → Mutiple: each uniquely identified by name/number in instr.
3. Immediate: The operand val is embedded in the instr. itself.
4. I/O device: specifies I/O module & device.

(For memory Mapped I/O, treated like memory address.

Instruction Representation





Binary Format:

⊙ Each instn is a seqn of bits divided into fields (opcode, operand etc)

⇒ The proc loads the into its IR to extract the required fields.

Symbollic Rep: intermediary in programming

complex to read binary code.

→ Mnemonics (ADD, SUB, MUL, DIV, LOAD, STORE) represent opcode.

→ Operands also symbollically rep

ADD R,Y means Add val at mem loc Y to reg R.

Q. Simple Prog asmts symbolic i.e

converts opcode & operand refs to binary form + constructs binary machine instr.

Instruction Types

Computational capabilities to process numeric data

1. Data processing: Arithmetic & logic instrs.
   - ↓ process any type of data user wants

2. Data storage: Movement of data to/from registers & mem loc.

3. Data movement: I/O instrs.

4. Control: Test & branch instr.

Instr. Set Design

1. Operation Repertoire: how many & which

operation to provide + their complexity.

2. Data Type:

3. Instr format: length (bits)
   # of address (0, 1, 2, 3)
   size of fields - - -

4. Register: How many + their use.

5. Addressing Mode: (specified operand)

Operand Types

Number
Logical data
Char
Address (unsigned int) [calc whether main/virtual memory.

Data Transfer

Data Transfer

Arithmetic

logical

Conversion

I/O

System Control

Transfer of Control

Data Transfer

↳ Location of source + destination specify.

→ length of data to be tranferred.

→ Addressing Mode.